

Hana Fujimoto

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Machine Learning Engineer (formerly civil engineering)

SKILLS

python, pytorch, onnx, opencv, cuda, numpy, ray, mlflow, polars

EXPERIENCE

Machine Learning Engineer | Sakura VisionOsaka, Japan | 04/2023 – Present

- Ships defect-detection models to 40 factory edge boxes across seven client sites in Kansai and Chubu.
- Cut inference cost 60% by exporting to ONNX and quantising to INT8, with a calibration set per line.
- Owns the retraining loop: Ray for sweeps, MLflow for the registry, and a shadow-deploy gate before promotion.
- Rewrote the data layer on Polars; nightly aggregation went from 50 minutes to under four.
- Wrote the drift monitor that catches lens fouling before the false-positive rate reaches the operators.

Computer Vision Engineer | Rinku RoboticsOsaka, Japan | 01/2022 – 03/2023

- Built the bin-picking pose estimator in PyTorch.

Machine Learning Intern | Naniwa Data LabOsaka, Japan | 07/2021 – 12/2021

- Six months part-time alongside the bridge role, which is how the transition was funded.
- Built the baseline segmentation model later open-sourced as the bridge crack dataset.
- Presented weekly to a reading group of nine engineers on papers from CVPR and MVA.

Structural Engineer | Kansai Bridge & StructureKobe, Japan | 04/2014 – 12/2021

- Load assessment for 60 prefectural road bridges under the 2016 revision of the Specifications for Highway Bridges.
- Led the seismic retrofit design for the Muko River crossing, a 340m continuous steel box girder.
- Ran finite-element models in Abaqus for fatigue life on orthotropic decks; published the calibration set.
- Supervised four junior engineers and the drawing office through two prefectural audits.
- Wrote the Python tooling that started the career change: a crack-width measurement pipeline over inspection photos that replaced two days of manual tracing per bridge.
- Standardised the inspection photo archive, 240k images, which later became the training set at Rinku.
- Represented the firm on the prefectural committee for bridge asset management, 2019-2021.

EDUCATION

MEng Civil Engineering, Kyoto University | Apr 2012 – Mar 2014

Certificate in Applied Statistics, Osaka Institute of Technology | Apr 2021 – Dec 2021

BEng Civil Engineering, Kyoto University | Apr 2008 – Mar 2012

PROJECTS

Bridge crack segmentation pytorch, opencv

Open dataset and baseline; 1.1k GitHub stars.

CERTIFICATIONS

Professional Engineer, Civil (Japan) — JSCE, Mar 2018

Deep Learning Specialisation — Coursera, Sep 2021

JDLA Deep Learning for Engineers — JDLA, Jul 2022

PUBLICATIONS

Crack width estimation on orthotropic steel decks from inspection photography. Journal of Bridge Engineering, Jun 2021

Transfer learning from civil inspection imagery to industrial defect detection. MVA, Jul 2023

A calibration set for INT8 export of small segmentation models. workshop paper, Nov 2024